

Zia Hasan, PhD
Postdoctoral Fellow, Neurosciences

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Academic Background:

- **Memorial University of Newfoundland (MUN) | St. John's, NL**
 - Ph.D. in Neurosciences (**cGPA: 4.0/4.0**) | Courses: Systems Neuroscience, Biostatistics | Feb 2026
- **University of Rajshahi | Rajshahi, Bangladesh**
 - M.S. in Genetic Engineering & Biotechnology (**cGPA: 3.71/4.0**) | Dec 2017
 - B.Sc. in Genetic Engineering & Biotechnology (**cGPA: 3.75/4.0**) | Mar 2015

Academic & Professional Experiences

- Leadership: Executive Director of Communications, Graduate Students' Union (GSU), MUN (Elected) 05/2022 - 04/2023
- Training visit to Dr. Kanae Ando's laboratory, Tokyo Metropolitan University, Japan 11/2022 – 01/2023
- Leadership: Executive Director of Communications, Graduate Students' Union (GSU), MUN (Elected) 05/2021- 04/2022
- Teaching Assistantships: Course: BIOL 2041-001, MUN 01/2021- 04/2021
- Teaching Assistantships: Course: BIOL 1001, MUN 09/2020- 12/2020
- Executive (Research & Development), Apex Biotechnology Laboratory (ABL), Bangladesh 11/2018 - 08/2019
- Research Fellow, Ministry of Science and Technology, Bangladesh 2016 - 2017
- Research Assistant (part-time), Joarder DNA & Chromosome Research Lab, University of Rajshahi, Bangladesh 2014 - 2016

Fellowships & Awards

- **Fellow of the School of Graduate Studies** | *Graduate Studies, MUN* | 2025-2026
- **Best PhD Oral Presentation (3rd, Neurosciences)** | **CAS\$125** | *Faculty of Medicine, MUN* | 2025
- **Apu Akbar Memorial Award** | **CAS\$500** | *School of Graduate Studies, MUN* | 2023
- **ARC-NL Fellowship** | **CAS\$20,000** | *Aging Research Centre, NL* | 2020-2022
- **Dean's Building a Healthy Tomorrow Award** | **CAS\$2,544** | *Faculty of Medicine, MUN* | 2021-2022
- **Dr. Angus J. Neary Genetics Scholarship** | **CAS\$2,831** | *Faculty of Medicine, MUN* | 2020-2021
- **Dean's Fellowship Award** | **CAS\$80,000** | *Faculty of Medicine, MUN* | 2019-2023
- **NST Fellowship** | **CAS\$1000** | *Ministry of Science and Technology, Bangladesh* | 2016-2017
- **Talent Scholarship (Tuition Waiver)** | *University of Rajshahi, Bangladesh* | 2015-2016

Publications (Selected)

I. Articles published in peer-reviewed journals

- [1] **Hasan, Z.**, Torraville, S. E., Omoluabi, T., Maziar, A., Belolise, O. N., MacGowan, L. A., Flynn, C. M., Reinhardt, C., Walling, S. G., Benoukraf, T., & Yuan, Q. (2025). *Sex and life experience shape locus*

- coeruleus pretangle tau pathology*. *Alzheimer's & Dementia*, 22(3), e71285. <https://doi.org/10.1002/alz.71285>
- [2] Rae, R. J., Alberhasky, J. M. H., Baillet, M., Bangasser, D. A., Belloy, M. E., Berry, A. S., ...**Hasan, Z.**,... & Kelberman, M. A. (2026). Sex differences in neuromodulatory subcortical systems and their implications for Alzheimer's disease. *Alzheimer's & dementia*, 22(3), e71291. <https://doi.org/10.1002/alz.71291>
- [3] Sepahvand, T., Qin, T., **Hasan, Z.**, Power, K. D., Carew, S. J., Nazari, N., Janes, A. M., et al. (2025). *Second-order threat conditioning in the amygdala–posterior piriform cortex network*. *Communications Biology*, 8(1), 1–18. <https://doi.org/10.1038/s42003-025-08287-2>
- [4] Omoluabi, T. (†), **Hasan, Z.** (†), Piche, J. E. (†), Flynn, A. R. S., Doré, J. J. E., Walling, S. G., et al. (2025). *Locus coeruleus vulnerability to tau hyperphosphorylation in a rat model*. *Aging Cell*, 24(3), e14405. <https://doi.org/10.1111/accel.14405> † Co–first authors
- [5] Crossley, Q. Y. C., Omoluabi, T., Torraville, S. E., **Hasan, Z.**, et al. (2024). *Hippocampal hyperphosphorylated tau-induced deficiency is rescued by L-type calcium channel blockade*. *Brain Communications*, 6(2). <https://doi.org/10.1093/braincomms/fcae096>

II. Non-peer-reviewed articles/presentations

- [1] **Hasan, Z.**, Benoukraf, T., & Yuan, Q. (2026). *Updating the Reference Genome from mRatBN7.2 to GRCr8 Improves Single-Nucleus RNA-Sequencing Sensitivity in the Rat Brain* (Manuscript in Preparation).
- [2] **Hasan, Z.**, Benoukraf, T., & Yuan, Q. (2026). Sex and Life Experience Converge on Hippocampal Transcriptomic Responses in a Locus Coeruleus Pretangle Tau Model *Alzheimer's Association International Conference (AAIC)*; London, UK.
- [3] **Hasan, Z.**, Benoukraf, T., & Yuan, Q. (2025). Nuclei Isolation Protocol from Fresh and Frozen Tissue for Single-Nucleus RNA-Seq and ATAC-Seq. *protocols.io*. <https://dx.doi.org/10.17504/protocols.io.14egn228zg5d/v1>
- [4] **Hasan, Z.**, Torraville, S.E., Reinhardt, C., Benoukraf, T., & Yuan, Q. (July 2025). Life Experience Tunes Cell Type-Specific Transcriptomic Responses to Early Tau Pathology. *BioMedicine Graduate Symposium*; Memorial University, St. John's (NL), Canada.
- [5] **Hasan, Z.**, Torraville, S. E., Omoluabi, T., Maziar, A., Belolise, O. N., MacGowan, L. A., Flynn, C. M., Benoukraf, T., & Yuan, Q. (July 2025). Environmental modulation of early tau pathology in Alzheimer's disease: A pretangle-tau rat model. *Alzheimer's Association International Conference (AAIC)*; Toronto (ON), Canada.
- [6] **Hasan, Z.**, Torraville, S.E., Reinhardt, C., Benoukraf, T., & Yuan, Q. (May 2025). Postnatal stress exacerbates, while enrichment mitigates, locus coeruleus pretangle tau-induced gene alterations in the hippocampus. *Canadian Neuroscience Meeting (CAN 2025)*; Toronto (ON), Canada.
- [7] **Hasan, Z.**, Omoluabi, T., Piche, J. E., et al. (2025). Locus coeruleus vulnerability to tau hyperphosphorylation in a rat model. NCBI Gene Expression Omnibus (GEO). Accession: GSE272794.
- [8] **Hasan, Z.**, Torraville, S.E., Reinhardt, C., Benoukraf, T., & Yuan, Q. (November 2023). The effect of environmental factors in a rat pretangle tau model of Alzheimer's disease: A single-nuclei transcriptomics study. *Society for Neuroscience (SfN) Annual Meeting*; Washington (DC), USA.
- [9] **Hasan, Z.** (July 2023). The effects of early life events in a rat pretangle tau model: a single-nuclei transcriptomics study. *BioMedicine Graduate Symposium*; Memorial University, St. John's (NL), Canada.

Additional

- **Training visit:** Dr. Kanae Ando's laboratory, Tokyo Metropolitan University, Japan (11/2022 – 01/2023).